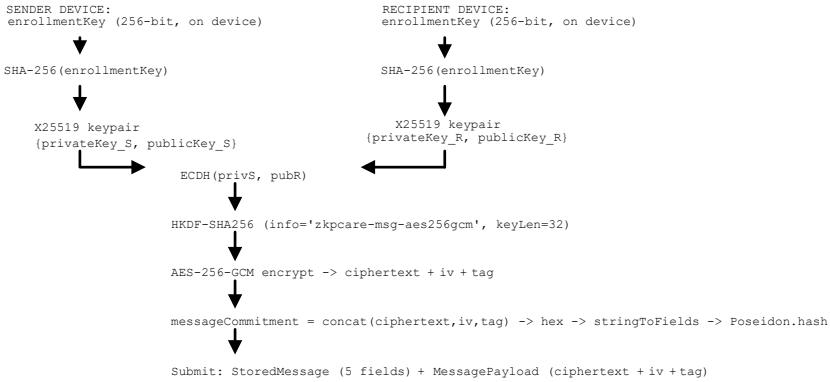
FIG. 11 -- ENCRYPTED MESSAGE STORE 1100

SECTION A: ENCRYPTION MODEL & KEY DERIVATION 1100



Key derivation is DETERMINISTIC: same enrollmentKey -> same keypair -> all past messages decryptable

SECTION B: MERKLE TREE(20) STRUCTURE 1102

```
          [ Root ]
            / \
         [H_L] [H_R]
     ... (20 levels) ...
  [lf0][lf1][lf2]...[lf_1M]
Capacity: 2^20 = 1,048,576 msgs
```

```
StoredMessage (5-Field Poseidon Hash):
 leafHash = Poseidon.hash([
  F1: messageCommitment (hash of ciphertext),
  F2: senderWalletId,
  F3: recipientWalletId,
  F4: messageType (1=clinical,2=admin,3=eScript),
  F5: threadIdHash
  ])
   Integrity in tree. Confidentiality off tree.
```

SECTION C: IN-TREE vs OFF-TREE SPLIT 1104

IN TREE (Poseidon leaf):		OFF TREE (stored alongside):
StoredMessage {		MessagePayload {
messageCommitment,		ciphertext (hex),
senderWalletId,		iv (hex),
recipientWalletId,		tag (hex),
messageType,		timestamp,
threadIdHash		receiptId?
} -> Poseidon.hash		} -> NOT hashed
= INTEGRITY		= CONFIDENTIALITY

SECTION D: FOUR PARALLEL INDEX STRUCTURES 1106

INDEX 1: senderIndex
Map<senderWalletId, bigint[]>

```
sender_A -> [0, 3, 5]
sender_B -> [2, 4]
```

INDEX 2: recipientIndex
Map<recipientWalletId, bigint[]>

```
recip_B -> [0, 2, 5]
recip_A -> [1, 3]
```

INDEX 3: walletIndex (UNION)
Map<walletId, bigint[]>

```
wallet_A -> [0, 1, 3, 5]
wallet_B -> [0, 2, 4, 5]
```

INDEX 4: threadIndex
Map<threadIdHash, bigint[]>

```
thread_T1 -> [0, 2, 5]
thread_T2 -> [1, 3, 4, 6]
```

6 RPC: l2_getMessagesByWallet, l2_getMessagesBySender, l2_getMessagesByRecipient,
l2_getMessagesByThread, l2_getMessageStoreRoot, l2_storeMessage

SECTION E: NFC RECOVERY FLOW 1108 BATCH INTEGRATION 1110:

L2Batch { messageStoreRoot?: string }

1. Tap NFC Card -> PIN decrypt -> enrollmentKey recovered Snapshot of tree root at batch
2. enrollmentKey -> SHA-256 -> X25519 keypair (same as before) creation, committed alongside
3. l2_getMessagesByWallet(walletId) -> encrypted entries claimAccumulatorRoot.
4. AES-256-GCM decrypt locally -> full history restored

Neither NFC card alone nor sequencer alone can read messages. Both needed.